

Methodology

Methodology I

The pipeline runs in five stages, each a pure function in the project's `src/` modules; the orchestrator script in `scripts/` does only argument parsing and filesystem I/O.

Read I

`infrastructure.prose.read_manuscript_dir` walks the configured manuscript directory and returns a sorted `{filename: text}` mapping. Scaffolding files (`preamble.md`, `config.yaml.example`, `AGENTS.md`, `README.md`, `SYNTAX.md`) are excluded by default — they hold infrastructure documentation rather than manuscript prose.

Analyse I

For each file, three analysers run in parallel:

- ▶ **Metrics** (`infrastructure.prose.analysis.metrics`) — word, sentence, paragraph, syllable counts; average words per sentence; average syllables per word; complex-word fraction; Flesch Reading Ease [`@flesch1948new`]; Flesch-Kincaid Grade Level [`@kincaid1975derivation`]; Gunning Fog Index [`@gunning1952technique`]. The implementations are textbook formulae over a vowel-group syllable heuristic; they're good enough for a writer-facing signal, not for linguistic research.
- ▶ **Structure** (`infrastructure.prose.analysis.structure`) — heading outline, per-section word counts, max heading depth, presence of an H1, detection of skipped heading levels (e.g. an H1 followed directly by an H3).

Analyse II

- ▶ **Quality** (`infrastructure.prose.analysis.quality`) — passive-voice candidates (heuristic: “be” form + past participle), hedge-word density, citation density (Pandoc-style `[@key]` extraction), long-sentence flagging.

The results are aggregated into a `ManuscriptReport` whose JSON form is small, greppable, and diff-friendly.

Cross-check I

Every cited key is matched against the BibTeX file at `bibliography.references_path`. The check has four tunable behaviours via `config.yaml`:

Setting	Effect
<code>fail_on_missing: true</code>	Any cited <code>[@key]</code> that does not exist in the bib fails the check.
<code>fail_on_missing: false</code>	Missing citations are warned but do not fail the run.
<code>fail_on_unused: true</code>	Bib entries that are never cited fail the check.
<code>fail_on_unused: false</code>	Unused entries are warned but do not fail.

Cross-check II

The check uses `infrastructure.reference.citation.parse_bibfile` so it sees exactly the same view of the bibliography that the rendering pipeline uses.

Evaluate I

The report runs through a set of pure check functions:

- ▶ `_check_grade_level` — the manuscript-wide weighted average FKGL falls within `[target_grade_level_min, target_grade_level_max]`.
- ▶ `_check_citation_density` — citations per 1000 words `citation_density_min_per_1000`.
- ▶ `_check_no_skipped_levels` — no file uses a skipped heading level (when `prose.forbid_skipped_levels: true`).
- ▶ `_check_h1_per_file` — every file has an H1 (when `prose.require_h1_per_section: true`).
- ▶ `_check_bibliography` — citation/bib consistency per the `bibliography: policy` block.

Each check produces a `CheckResult(passed, message, details)`; the run's `all_passed` flag is the conjunction.

Render I

`src/report.py::write_review_report` writes a markdown file with:

- ▶ Top-line counts (files, words, sentences, paragraphs, averages).
- ▶ A pass/fail table for every check.
- ▶ Per-file metrics table (word count, sentence count, FRE, FKGL, Fog, citations, hedges, passives).
- ▶ Heading outline per file.
- ▶ Quality-flag callouts (long sentences, passive candidates, hedges) per file.

Three orchestrators run in lexicographic order (so the figure and variable stages always see a fresh review):

Render II

- ▶ `scripts/run_prose_pipeline.py` writes `output/manuscript_report.json` (the raw `ManuscriptReport`), `output/checks.json` (the list of check results), `output/review_report.md` (the human-readable review report), and `output/run_summary.json` (one-line metadata).
- ▶ `scripts/y_generate_prose_figures.py` reads `manuscript_report.json` and writes `output/figures/{section_word_counts,readability_metrics}`
- ▶ `scripts/z_generate_manuscript_variables.py` reads `manuscript_report.json`
 - ▶ `config.yaml` and writes `output/data/manuscript_variables.json` along with a resolved manuscript tree under `output/manuscript/`.
- ▶ `manuscript/references.bib` is **not** modified by this pipeline — the prose project does not generate citations, only validates them.